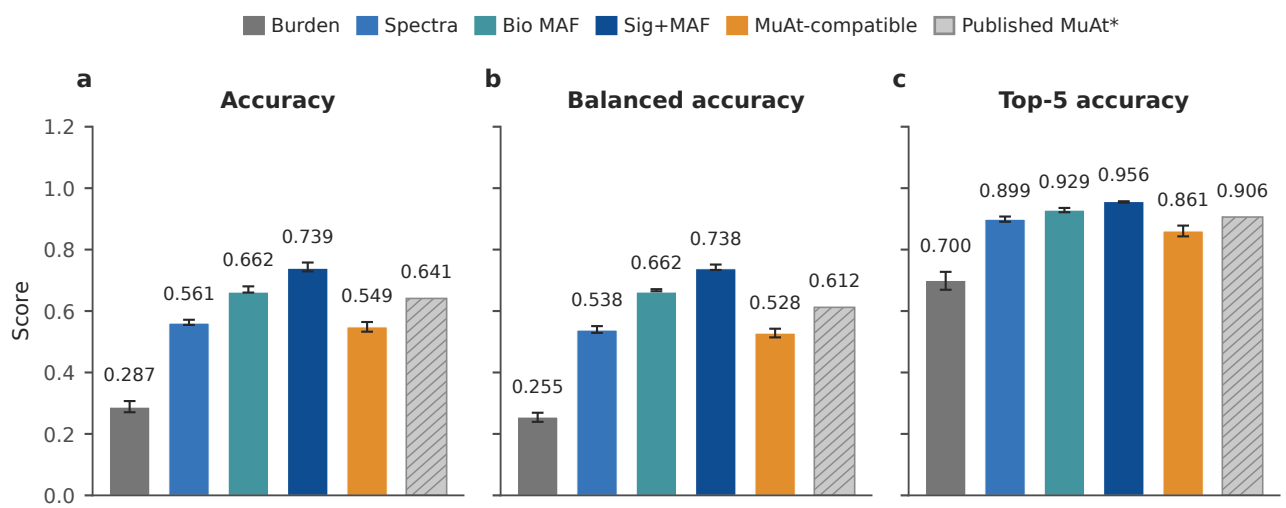

** Published MuAt: contextual (n = 7,352, official framework, ten-model ensemble); balanced accuracy 0.612 derived from the paper's supplementary per-class recall (Table S2).*